Chapter 1 : Examining Distributions

31 May 2024 16:37



Prelude

The case of the missing vans

Auto manufacturers lend their dealers money to help them keep vehicles on their lots. The loans are repaid when the vehicles are sold. A Long Island auto dealer named John McNamara borrowed over $6 billion from General Motors between 1985 and 1991. In December 1990 alone, Mr. McNamara borrowed $425 million to buy 17,000 GM vans customized by an Indiana company for sale overseas. GM happily lent McNamara the money because he always repaid the loans.

Let's pause to consider the numbers, as GM should have done but didn't. The entire van-customizing industry produces only about 17,000 customized vans a month. So McNamara was claiming to buy an entire month's production. These large, luxurious, and gas-guzzling vehicles are designed for U.S. interstate highways. The recreational vehicle trade association says that only 1.35% were exported in 1990. It's not plausible to claim that 17,000 vans in a single month are being bought for export. McNamara's claimed purchases were large even when compared with total production of vans. Chevrolet, for example, produced 100,047 full-sized vans in all of 1990.

Having looked at the numbers, you can guess the rest. McNamara admitted in federal court in 1992 that he was defrauding GM on a massive scale. The Indiana company was a shell set up by McNamara, its invoices were phony, and the vans didn't exist. McNamara borrowed vastly from GM, used most of each loan to pay off the previous loan (thus establishing a record as a good credit risk), and skimmed off a bit for himself. The bit he skimmed amounted to over $400 million. GM set aside $275 million to cover its losses. Two executives, who should have looked at the numbers relevant to their business, were fired.[1]

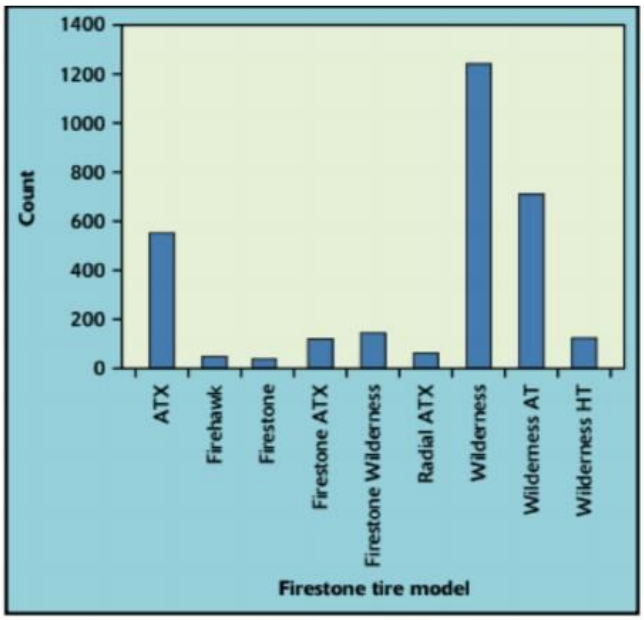
⇒ Below are Two basic strategy to help organize the EDA.

1) Begin by examining each variable by itself. Then move on to study the relationship among the variable.

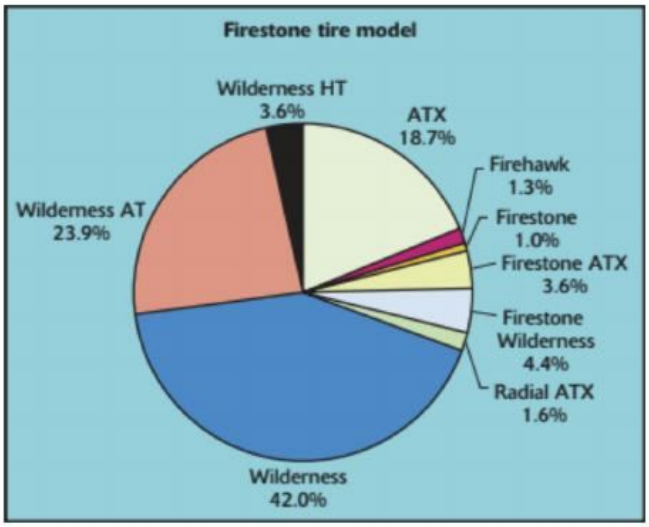
2) Begin with graphs, and then add numerical summaries of specific aspects of data.

° Categorical variables: Bar graphs and pie charts :-

⇒ The bar graph quickly compares the size of each category.



⇒ The pie chart helps us see what part of whole each group forms.



- Bar graphs are more flexible than pie-chart :-
  - ex:- You can use bar chart to compare numbers of student taking science and commerce.
    - Not all students fall in these two category (some may take arts for example). A pie-chart hence can't make this comparision without slight modification.

- A bar graph whose categories are arranged in order of the percentages (highest to lowest) is called a <u>pareto chart</u>.

- <u>State Unemployment Rates</u> :-

TABLE 1.1 Unemployment rates by state, December 2000

State	Percent	State	Percent	State	Percent
Alabama	4.0	Louisiana	5.3	Ohio	3.7
Alaska	6.1	Maine	2.6	Oklahoma	2.6
Arizona	3.3	Maryland	3.3	Oregon	4.0
Arkansas	3.9	Massachusetts	2.0	Pennsylvania	3.8
California	4.3	Michigan	3.4	Puerto Rico	8.9
Colorado	2.1	Minnesota	2.8	Rhode Island	3.2
Connecticut	1.5	Mississippi	4.3	South Carolina	3.3
Delaware	3.3	Missouri	3.2	South Dakota	2.3
Florida	3.2	Montana	4.9	Tennessee	3.8
Georgia	3.0	Nebraska	2.5	Texas	3.4
Hawaii	3.6	Nevada	4.0	Utah	2.7
Idaho	5.0	New Hampshire	2.2	Vermont	2.4
Illinois	4.5	New Jersey	3.5	Virginia	1.9
Indiana	2.7	New Mexico	4.9	Washington	4.9
Iowa	2.5	New York	4.2	West Virginia	5.5
Kansas	3.2	North Carolina	3.6	Wisconsin	3.0
Kentucky	3.7	North Dakota	2.7	Wyoming	3.7

Source: Bureau of Labor Statistics.

⇒ We'll examine the above data in detail, starting with the nature of variable recorded.

Q) What does it mean to be unemployed?

⇒ To be unemployed, a person must first be in the

labour force, i.e the must be available for work
↳ looking for work.

$$\text{unemployment rate} = \frac{\text{number of people in labour force that are unemployed}}{\text{number of people in labour force}}$$

⇒ To understand the unemployment situation, we begin with graph.

□ A histogram of state unemployment rates :-

a) To make a histogram of the data,

b) Divide the range of data into classes of equal width. The unemployment rates ranges from 1.5% to 8.9%, so we choose our classes.

$1.0 \leq$ unemployment rate < 2.0
$2.0 \leq$ unemployment rate < 3.0
⋮
$8.0 \leq$ unemployment rate ≤ 9.0

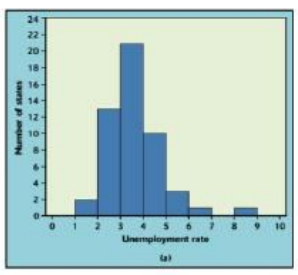
c) Count the number of individuals in each class.

Class	Counts
1.0 to 1.9	2
2.0 to 2.9	13
3.0 to 3.9	21
4.0 to 4.9	10
5.0 to 5.9	3
6.0 to 6.9	1
7.0 to 7.9	0
8.0 to 8.9	1

The bases of a histogram should cover the entire range of values of a variable, with no space between bases unless a class is empty.

↳ When the possible values of a variable have gaps between them

↳ extend the bases of the bars to meet halfway between two adjacent possible values.





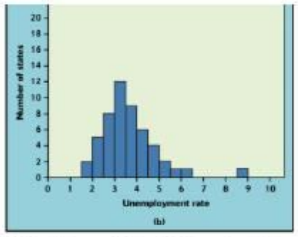


FIGURE 1.2 Histograms of the December 2000 unemployment rates in the 50 states and Puerto Rico, from Table 1.1. The two graphs differ only in the number of classes used.

⇒ Also having too many graph will produce a "pancake" graph, with most classes having one or no observations.

□ Interpreting histograms :-

1) In any graph of data, look for overall pattern and for striking deviations from that pattern.

2) You can describe the overall pattern of histogram by its shape, center and spread.

3) The important kind of deviation is an outlier, an individual value that falls outside the overall pattern.

EXAMPLE 1.4 **The distribution of unemployment rates**

CASE 1.1

Look again at the histogram in Figure 1.2(b). The first feature we notice is an *outlier*. Puerto Rico has much higher unemployment (8.9%) than any of the 50 states. Some outliers are due to mistakes, such as typing 5.9 as 8.9. Other outliers point to the special nature of some observations. Although Puerto Rico is included in the government report from which we took the unemployment data, it is not a state and we should omit it when thinking about domestic unemployment.

Shape: The distribution has a *single peak*. Even omitting the outlier, the histogram is somewhat *right-skewed*—that is, the right tail extends farther from the peak than does the left tail. The states with unemployment rates at or higher than 5% are Alaska, Idaho, Louisiana, and West Virginia. These states are often near the top in unemployment due to generally weak job opportunities. Note that winter weather does not explain high unemployment in Alaska and Idaho—the data are *seasonally adjusted* to compensate for regular changes such as job losses in the winter months in northern states. **Center:** The midpoint of the 51 observations is close to the single peak, at about 3.5%. **Spread:** The spread is 1.5% to 6.5% if we ignore Puerto Rico.

⇒ When you describe a distribution, concentrate on the main features.

- Look for major peaks, not for minor ups and downs in the bars of the histograms.
- Look for clear outliers, not just for smallest and largest observations.
- Look for rough symmetries or clear skewness.

○ Symmetric And Skewed Distributions :-

⇒ A distribution is symmetric if the right & left sides are approximately the mirror images of each other.

⇒ A distribution is skewed to the right if the right side of the histogram extends much farther out than the left side.

↳ opposite case for distributions skewed to the left.

- Quantitative variables : Stemplots :-

⇒ for a small data set, a stemplot is quicker to make and presents more detailed information.

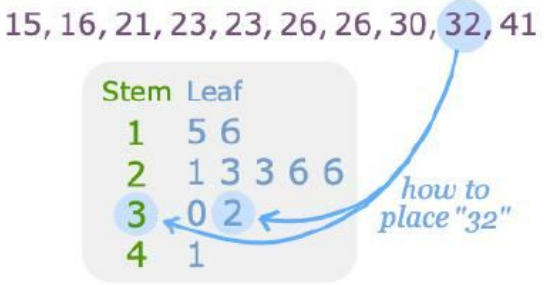
□ To make a stemplot

1. Seperate each observation into a stem consisting of all but the final (rightmost) digit and a leaf, the final digit.

↳ Stems can have as many digits as possible, but each leaf contains only single digit.

2. Write the stem in a vertical column with the smallest at the top, and draw the vertical line at the right of this column.

3. Write each leaf in the row to the right of its stem, in increasing order out from stem.



◈ When the observed values have many digits, it's often best to round the numbers to just a few digits before making a stemplot.

⇒ Stemplots work well for smaller data, when the number of data is greater than or equal to 100, an histogram is always better.

- Time Plots :-

⇒ Many variables are measured at intervals over time.

↳ ex:- cost of raw material each month, price of stock.

◈ In such variables our interest is change over time.

⇒ A time plot of a variable plots each observation against the time at which it was measured.

→ Always put the time on horizontal scale and the variable you are measuring on vertical scale (y-axis).

→ Connecting the datapoints via lines helps emphasize any change over time.

Measurements of variable over time forms a time series.

Seasonal Variations and trends :-

1) A pattern in time series that repeats itself at a known regular interval of time is called a seasonal variation.

2) A trend in a time series is a persistent, long term rise or fall.

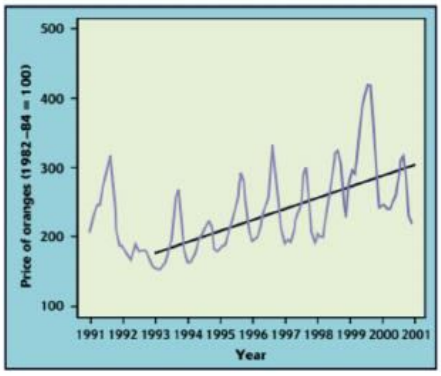


FIGURE 1.7 Time plot of an index number (1982–84 = 100) for the average retail price of oranges each month from 1991 through 2000, for Example 1.7.

Such a line appears in Figure 1.7. Superimposed on this trend is a strong *seasonal variation*, a regular rise and fall that recurs each year. Orange prices are usually highest in August or September, when the supply is lowest. Prices then fall in anticipation of the harvest and are lowest in January or February, when the harvest is complete and oranges are plentiful. The large spikes in 1991 and 1999 may be deviations from the overall trend and seasonal variation, perhaps due to poor harvests.

1.2 Describing Distribution with numbers :-

CASE 1.2

EARNINGS OF HOURLY BANK WORKERS

Banks employ many workers paid by the hour, such as tellers and data clerks. A large bank, which we will call simply National Bank to preserve confidentiality, has been accused of discrimination in paying its hourly workers. In the course of an internal investigation into the charges, the bank collected data on all 1745 hourly workers listed in its personnel records. The data set is *hourly.dat*, described in detail in the Data Appendix at the back of the book. Because the data set is so large, we will use randomly selected samples of full-time hourly workers in four race-and-gender groups to illustrate statistical analysis before applying the methods we develop to the complete data set. Table 1.8 presents the annual earnings for the workers in our samples.

TABLE 1.8 Annual earnings of hourly workers at National Bank

Black females	Black males	White females	White males
$16,015	$18,365	$25,249	$15,100
17,516	17,755	19,029	22,346
17,274	16,890	17,233	22,049
16,555	17,147	26,606	26,970
20,788	18,402	28,346	16,411
19,312	20,972	31,176	19,268
17,124	24,750	18,863	28,336
18,405	16,576	15,904	19,007
19,090	16,853	22,477	22,078
12,641	21,565	19,102	19,977
17,813	29,347	18,002	17,194
18,206	19,028	21,596	30,383
19,338		26,885	18,364
15,953		24,780	18,245
16,904		14,698	23,531
		19,308	
		17,576	
		24,497	
		20,612	
		17,757	

⇒ The below Stemplot shows us the shape, center, and spread of earning of 15 black females shown above.

↳ Stems are thousands of dollars, and the leaves are hundreds, rounded to the nearest hundred.

12	6
13	
14	
15	
16	0 0 6 9
17	1 3 5 8
18	2 4
19	1 3 3
20	8

⇒ As is often the case when there are few observations, the shape of distribution is irregular

- Measuring center: the mean

⇒ The most common measure of center is the ordinary arithmetic average or mean.

$$\therefore \text{Mean } \bar{x} = \frac{x_1 + x_2 + \dots + x_n}{n}$$

$$\bar{x} = \frac{1}{n}\sum x_i$$

□ Mean earning of black female workers:-

$$\bar{x} = \frac{x_1 + \dots + x_n}{n}$$

$$= \frac{16,015 + 17,516 + \dots + 16,904}{15}$$

$$= \frac{262,934}{15} = \$\,17,528.93.$$

⇒ The lowest paid worker earned $12,641. If this is excluded the mean becomes $\bar{x} = \$\,17,878.07$.

↳ The single outlier reduces the mean by almost $350.

Mean is sensitive to the influence of one or more extreme observations

A skewed distribution that has no outliers will also pull the mean towards its long tail.

Because the mean cannot resist the influence of extreme observation it is not a resistant measure of center

- Measuring center : the median

⇒ The median M is the midpoint of a distribution, the number such that half of the observations are smaller, and other half are larger. To find the median of a distribution :

1) Arrange all observations in order of size, from smallest to largest.

2) If the number of observation n is odd, the median is the center observation $\underbrace{(n+1)/2}_{\text{index}}$.

3) If the number of observations n is even, the median is the mean of two observation in the center.

⇒ In both 263 the location of median is again $\overbrace{(n+1)/2}^{\text{index}}$ from the bottom of the list.

- Mean earnings of black female workers.

⇒ first arrange the data in order from smallest to largest.

12641, 15953, 16015, 16555, 16904, 17124, 17274, 17516, 17813, 18206, 18405, 19090, 19312, 19338, 20788.

location of M ⇒ $\frac{n+1}{2}$ ⇒ $\frac{16}{2}$ ⇒ 8

∴ 17516 $, it has 7 observation to the left & 7 observation to the right.

Q) How much does the outlier $12,641 affect the median?

⇒ Drop it from the list and the median location will be :

$$\frac{n+1}{2} = \frac{15}{2} = 7.5 \text{ (between 7 \& 8).}$$

i.e mean of the 7th & 8th observation.

$$\therefore \text{Median} = \frac{17,516 + 17,813}{2}$$

$$= 17,664.50\ \$.$$

- Comparing Mean & Median :-

⇒ The low outlier pulls the mean by $350, while the median drops by only $148, when we add the low outlier to our list.

⇒ The median is more resistant than mean.

⇒ The mean and median of a symmetric distribution are close together. If the distribution is exactly symmetric, the mean & median are exactly the same.

⇒ In a skewed distribution, the mean is farther out in the long tail than is the median.

tail than is the median. For example, the distribution of house prices is strongly skewed to the right. There are many moderately priced houses and a few very expensive mansions. The few expensive houses pull the mean up but do not affect the median. The mean price of existing houses sold in 2000 was $176,200, but the median price for these same houses was only $139,000. Reports about house prices, incomes, and other strongly skewed distributions usually give the median ("midpoint") rather than the mean ("arithmetic average"). However, if you are a tax assessor interested in the total value of houses in your area, use the mean. The total is the mean times the number of houses, but it has no connection with the median. The mean and median measure center in different ways, and both are useful.

- Measuring spread : the quartiles :-

⇒ A measure of center alone can be misleading.

- If two nations with same median household income are very different if one has extremes of wealth & poverty, and other has little variations among households.

ex A drug with correct mean concentration of active ingredients is dangerous if some batches are too much high and others have too less concentration.

⇒ We're interested in spread or variability of income & drug potencies

as well as their centers.

⇒ The simplest useful numerical description of a distribution consist of both center and a measure of spread.

⇒ The single observation show full spread of the data, but they may be outliers.

- p^{th} percentile :-

⇒ The pth percentile of a distribution is the value such that [p Percent] of observations fall at or below it.

→ The median is just 50th percentile of the data

Use of percentile is particularly appropriate when the median is the measure of the center.

- Quartile

⇒ The most commonly used [percentile] other than median are the quartile. (is a)

⇒ The first quartile is 25th percentile, and the third quartile is 75th percentile.

→ i.e the first & third quartile shows the spread of middle half of the data.

→ The second quartile is the median itself.

- The quartiles Q_1 and Q_3 :-

⇒ to calculate the quartile.

1) Arrange the observations in increasing order, and locate the median M in the ordered list of observations.

2) The first quartile Q_1 is the median of the observations whose position in the ordered list is the left of the location of all over data. (median of the first half).

3) The third quartile Q_3 is the median of the observations whose position in the ordered list is the right of the location of all over data. (median of the second half).

- finding quartiles for the black female workers :-

12641 15953 16015 16555 16904 17124 17234 17516
17813 18206 18405 19090 19312 19338 20788

$n = 15 \Rightarrow$ odd $\Rightarrow \frac{n+1}{2} \Rightarrow 8^{th}$ observation is median (2nd Quartile) ($17,516)

Now, first quartile is median of first 7 obs.

$n = 7 \Rightarrow$ odd $\Rightarrow \frac{n+1}{2} \Rightarrow 4^{th}$ observation is Median a.k.a first quartile ($16,555 = Q_1)

⇒ And third quartile is the median of the last 7 observation.

$n = 7 \Rightarrow$ odd $\Rightarrow \frac{n+1}{2} \Rightarrow 4^{th}$ observation after the overall median is the 3rd Quartile. ($19,090 = Q_3)

Notice that quartiles are resistant (→ Not immune.), ex:- Q_1 would have same value if the low outlier 12,641 where 1000.

- The five-number summary and boxplots :-

⇒ To get a quick summary of both center and spread combine all five numbers : lowest_val, Q_1, M, Q_3, highest_val.
  - → The result is a five number summary and a graph based on it.

- Box plot is a graph of the five-number summary.
  - → A central box spans the quartiles.
  - → A line in the box marks the median.
  - → Lines extend from the box out to the smallest and largest observations.

Box plots are most useful for side by side comparision of several distribution

⇒ You can draw a boxplot horizontally or vertically. Be sure to include a numerical scale in graph.

⇒ When you look at a box plot.
  - → First locate the median, which marks the center of distribution.
  - → Then look at the spread, the quartiles show the spread of middle half of the data
  - → Then look at the extremes, shows the spread of entire data.

⇒ We now have the tools for a preliminary examination of the National Bank earnings data.

EXAMPLE 1.12 **Adverse impact at National Bank**

CASE 1.2

The complete data set *hourly.dat* contains information about 1276 full-time hourly workers at National Bank. Among these workers are 315 black females, 697 white females, 81 black males, and 162 white males. Statistical software gives us the five-number summaries for the earnings of each group:

	A	B	C	D	E	F	G	H
1	Earnings	N	Min	25%	Median	75%	Max	
2	Black female	315	12641	17145.5	18354	20363	33703	
3	White female	697	12641	18492	21492	25583	37513	
4	Black male	81	14731	17155	18400	20972	32678	
5	White male	162	12605	18343.25	22029.5	25302	36608	
6								

The boxplots in Figure 1.11 compare the four distributions of earnings. Concentrate on the box spanning the quartiles, because the extremes in a large data set are often outliers. We see at once that whites, both female and male, tend to earn more than blacks. The median income of white hourly workers is greater than the third quartile of black income. There are only minor differences between females and males within each race. In legal terms, the data demonstrate that National Bank's pay policies have an *adverse impact* on blacks. The bank must now investigate whether there is a legitimate reason for the black/white earnings difference, such as hours worked, seniority, or job classification.

⇒ The data demonstrates that the Bank's pay policies have an adverse impact on black.

↳ The bank must now investigate whether there is a legitimate reason for the difference such as hours worked, seniority, or job classification.

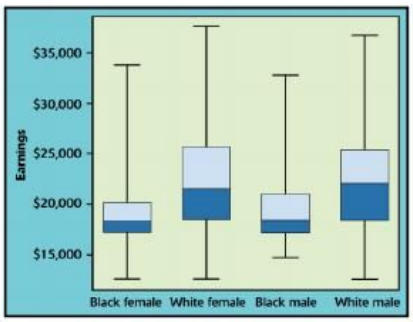


FIGURE 1.11 Side-by-side boxplots comparing the earnings of four groups of hourly workers at National Bank, for Example 1.12.

Figure 1.11 illustrates the power of simple statistical descriptions to make sense of a large amount of information. Although of less interest in this case, boxplots also suggest the symmetry or skewness of a distribution. In a symmetric distribution, the first and third quartiles are equally distant from the median. In most distributions that are skewed to the right, on the other hand, the third quartile will be farther above the median than the first quartile is below it. The extremes behave the same way, but remember that they are just single observations and may say little about the distribution as a whole, especially if either is an outlier. Figure 1.11 suggests that the earnings distributions are somewhat right-skewed.

Box plot also suggest the symmetry or skewness of distribution.

↳ In symmetric distribution the first and third quartile are equally distant from median.

↳ In right skewed distributions, the third quartile is further above the median than the first quartile below it.

Box plot identifying skewness

- Measuring the spread: the standard deviation :-

⇒ The standard deviation measures spread by looking at how far the observations are from their mean.

⇒ The variance s^2 of a set of observations is the average of the squares of deviations of the observations from their mean. In symbol the variance of n observations $x_1, x_2, \ldots, x_n$ is

$$s^2 = \frac{(x_1-\bar{x})^2 + (x_2-\bar{x})^2 + \ldots + (x_n-\bar{x})^2}{n-1}$$

where $\bar{x}$ = mean

or more compactly

$$s^2 = \frac{1}{n-1}\sum_{i=1}^{n}(x_i-\bar{x})^2$$

Sample variance with bessell correction.

Population variance $= \frac{1}{N}\sum_{i=1}^{N}(x_i-u)^2$

u = population.

⇒ The standard deviation is the square root of the variance s^2.

$$s = \sqrt{\frac{1}{(n-1)}\sum_{i=1}^{n}(x_i-\bar{x})^2}$$

- Calculating the standard deviation :-

ex:- The bureau of labour statistics lists average hourly wages for 8 categories of law related occupations (the units are dollars per hour).

30 17 36 14 17 12 15 17

1) Find the mean.

$$\bar{x} = \frac{30+17+36+14+17+12+15+17}{8}$$

$$= \frac{158}{8} = 19.75 \text{ \$ per hour.}$$

Figure 1.12 displays the data as points above the number line, with their mean marked by an asterisk (*). The arrows mark two of the deviations from the mean.

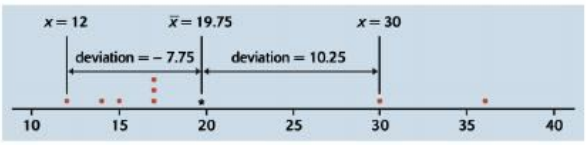


FIGURE 1.12 Average hourly wages for eight categories of legal professionals, with their mean (•) and the deviations of two observations from the mean, for Example 1.13.

The deviations show how spread out the data are about their mean. They are the starting point for calculating the variance and the standard deviation.

Observations x_i	Deviations $x_i - \bar{x}$	Squared deviations $(x_i - \bar{x})^2$
30	$30 - 19.75 = 10.25$	$10.25^2 = 105.0625$
17	$17 - 19.75 = -2.75$	$(-2.75)^2 = 7.5625$
36	$36 - 19.75 = 16.25$	$16.25^2 = 264.0625$

14	$14 - 19.75 = -5.75$	$(-5.75)^2 = 33.0625$
17	$17 - 19.75 = -2.75$	$(-2.75)^2 = 7.5625$
12	$12 - 19.75 = -7.75$	$(-7.75)^2 = 60.0625$
15	$15 - 19.75 = -4.75$	$(-4.75)^2 = 22.5625$
17	$17 - 19.75 = -2.75$	$(-2.75)^2 = 7.5625$
	sum = 0	sum = 507.5000

⇒ The variance is sum of the squared deviations divided by 1 less than the number of observations

$$s^2 = \frac{507.5}{7} = 72.5$$

⇒ The standard deviation is the square root of variance.

$$s = \sqrt{72.5} = 8.5147 \text{ \$ per hour.}$$

Degrees of freedom :-

⇒ Notice the "average" is variance s^2 divides the sum by 1 less than the number of observations.

⇒ The reason is that the deviations $x_i - \bar{x}$ always sum to 0.

→ Proof $\sum_{i=1}^{n}(x_i - \bar{x}) = \sum_{i=1}^{n} x_i - n\bar{x}$

$= n\bar{x} - n\bar{x}$

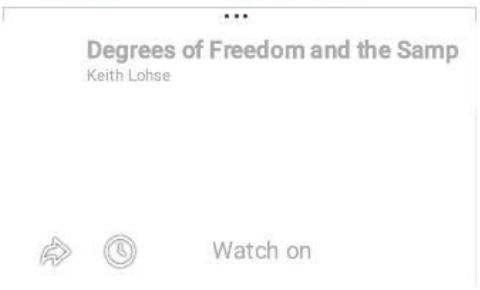
$= 0$.

→ So knowing only $n-1$ of the deviations can determine the last one.

→ Only $n-1$ of squared deviations can vary freely.

Degree of freedom of an estimate is the number of independent pieces of information on which the estimate is based.

Degrees of Freedom and the Sample Variance



⇒ Properties that determines the useful of the standard deviation.

0 s measures the spread around mean & should only be used when the mean is chosen as the measure of center.

2) $s=0$ only when there is no spread. This happens only when all observations have same value. Otherwise $s>0$.

3) s has same units of measurements as the original observations.

4) like the mean $\bar{x}$, s is not resistant.

⇒ standard deviation is the natural measure of spread for an important class of symmetrical distributions ⇒ the normal distributions

The five number summary is usually better than the mean and standard deviation for describing a skewed distribution or a distribution with extreme outliers.

↳ Use $\bar{x}$ and s only for reasonably symmetric distributions that are free of outliers.

□ Risk and Return

⇒ It's usual in finance to measure risk by the standard deviations of returns.

↳ On the ground that investments whose returns show larger spread are risky.

⇒ Remember a graph gives the best overall picture of a distribution

↳ Numerical measures of center and spread reports specific facts about a distribution, but they do not describe its entire shape.

o The Normal Distributions :-

1.3

⇒ the 4th step to EDA strategy:-

4) Sometimes the overall pattern of a large number of observations is so regular that we can describe it by smooth curve.

□ Density Curves :-

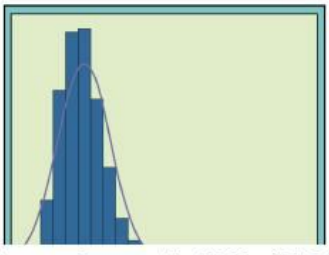




FIGURE 1.14 Histogram of the city gas mileage (miles per gallon) of 856 2001 model year motor vehicles. The smooth curve shows the overall shape of the distribution.

⇒ The figure above is a histogram of city gas mileage achieved by all 856 2001 model year motor vehicles listed in government's annual fuel economy report.

(model number)

⇒ The distribution is quite regular. With the exception of few outliers, the histogram is roughly symmetric.

↳ And both tails falls off quite smoothly from a single center peak.

⇒ The smooth curve overlying the histogram bars is an approximate description of the overall pattern of the data.

↳ It's a mathematical model for the distribution.

○ From Histogram to Density Curve :-

⇒ The bar areas in a histogram represent proportions of the observations.

⇒ The area of shaded bar in Figure 1.15(a) represents the proportion of 2001 model year vehicles with mile per gallons rating of less than 20.

↳ There are 384 such vehicles, which represents 384/856 = 0.449 = 44.9% of all 2001 model year vehicles.

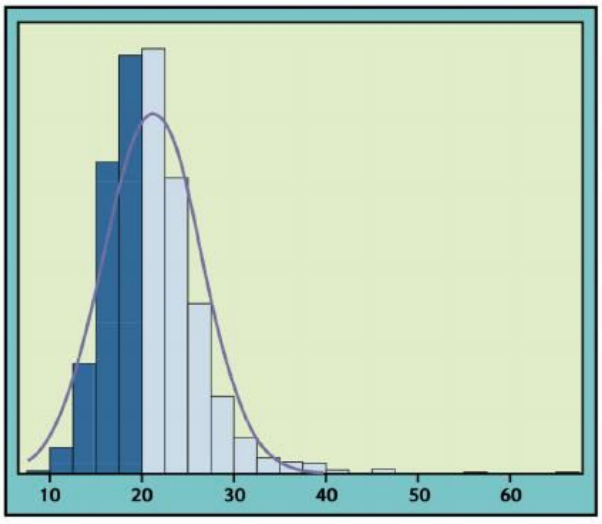


FIGURE 1.15(a) The proportion of gas mileages less than 20.0 from the histogram is 0.449.



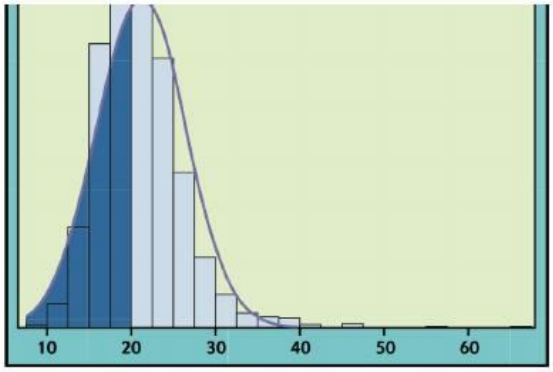


FIGURE 1.15(b) The proportion of gas mileages less than 20.0 from the density curve is 0.410.

⇒ In figure 1.15(b), the area under the curve to the left of 20 is shaded.

→ Adjust the scale of the graph, so that the total area under the curve is exactly 1. (i.e the proportion 1 represents all observations, so can't increase further).

→ This curve is now a density is now a density curve.

⇒ The area of proportion of 2001 model year vehicles with miles per gallon rating of less than 20 is 0.410

→ Only 0.039 away from histogram

□ Density curve definition :-

⇒ A density curve is a curve that
1) Is always on, or above horizontal axis, and
2) has area exactly 1 underneath it.

⇒ Density curves come in many shapes (not just normal!).

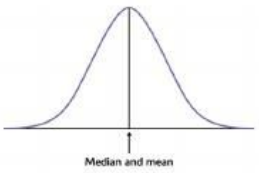


FIGURE 1.16(a) The median and mean of a symmetric density curve.

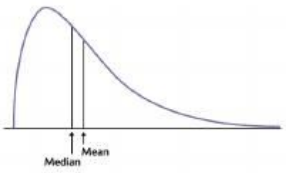


FIGURE 1.16(b) The median and mean of a right-skewed density curve.

• Median & Mean of a density curve :-

Median: The median of a density curve is the equal-areas point, the point that divides the area under the curve in half.

Mean: The mean of a density curve is a balance point, at which the curve would balance if made of solid material.

⇒ The median and mean are the same for a symmetric density curve. They both lie at the center of curve. The mean of a skewed curve is pulling away from median in direction of long tail.

→ The usual notation of idealized distribution is for mean = μ, and standard deviation = σ.

- Normal Distributions :-

⇒ These density curves are symmetric, single-peaked and bell-shaped.

⇒ All normal distributions have the same overall shape.

⇒ The exact density curve of a particular normal distribution is described by the mean μ and its standard deviation σ.

⇒ The mean & median are located at the center (same spot) of the symmetric curve.

⇒ Changing the μ without changing σ moves the normal curve along the horizontal axis without changing its spread.

↳ Only the standard deviation σ controls the spread.

- The 68-95-99.7 rule :-

⇒ In the N.D with mean μ and the standard deviation σ:

→ 68% of the observations fall within σ of the mean μ.
→ 95% of the observation fall within 2σ of μ.
→ 99.7% of the observations fall within 3σ of μ.

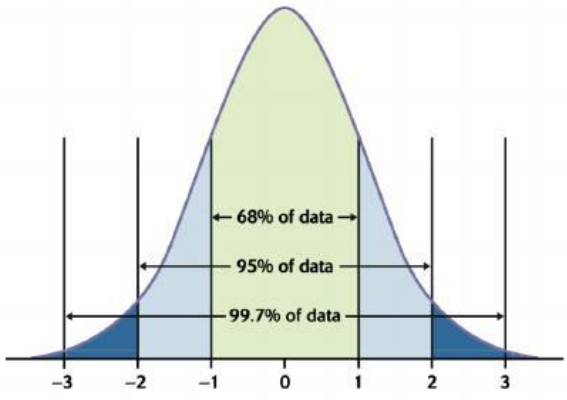


FIGURE 1.21 The 68–95–99.7 rule for Normal distributions.

⇒ We abbreviate the N.D with mean μ and standard deviation σ as $N(\mu, \sigma)$.

- The standard Normal Distribution :-

⇒ All Normal distributions are the same if we measure in units of size σ about the mean μ as the center.

↳ Changing to these units is called standardizing.

- Standardizing and z-scores :-

⇒ If x is an observation from a distribution that has a mean μ and standard deviation σ, the standardized value of x is

$$z = \frac{x-\mu}{\sigma} \quad \text{z-score}$$

A standardized value is often called a z-score.

A z-score tells us how many standard deviations the original observations falls away from the mean, and in which direction.

→ Observation larger than mean are +ve after standardization.
Observation smaller than mean are -ve after standardization.

- Example: Standardising the potato chip bags of weights.

⇒ The weight of 9-ounce potato chip bags are approximately Normal with $\mu = 9.12$ ounces, and $\sigma = 0.15$ ounces. The standardized weight is

$$z = \frac{\text{weight} - 9.12}{0.15} = 1.2$$

Or 1.2 standard deviations above the mean.

- Standard Normal Distribution : Definition.

⇒ The standard Normal Distribution $N(0, 1)$ with mean 0, and standard deviation 1. (0: mean, 1: std)

⇒ If a variable x has any Normal distribution $N(\mu, \sigma)$ with mean μ, and a standard deviation σ, then the standardized variable

$$z = \frac{x-\mu}{\sigma}$$

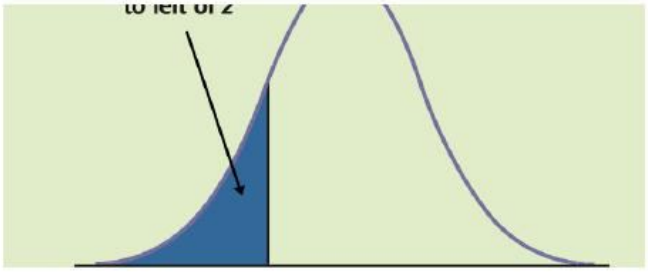
has the standard Normal distribution.

- Normal distribution calculations :-

⇒ Because all Normal distributions are the same when we standardize, we can find the areas using a single table.

⇒ The table of areas under the standard Normal curve has the table entry for each value z (z-score) is the area under the curve to the left of z.





⇒ We can answer any question about the proportion of observations in a N·D dist by standardizing and then using the standard ND table

1) State the problem in terms of the observed variable x.

2) Standard x to restate the problem in terms of a standard Normal variable z.
↳ Draw a picture → shade the area under the standard Normal curve.

3) Find the required area under the standard Normal curve using the tables and the fact that the total area under the curve is 1.
↳ i.e if you want to find the area to the right, just do 1 − area to the left.

EXAMPLE 1.20 **Normal distribution calculations**

The annual rate of return on stock indexes (which combine many individual stocks) is approximately Normal. Since 1945, the Standard & Poor's 500-stock index has had a mean yearly return of about 12%, with a standard deviation of 16.5%. Take this Normal distribution to be the distribution of yearly returns over a long period. The market is down for the year if the return on the index is less than zero. In what proportion of years is the market down?

1. *State the problem.* Call the annual rate of return for Standard & Poor's 500-stock index x. The variable x has the $N(12, 16.5)$ distribution. We want the proportion of years with $x < 0$.

2. *Standardize.* Subtract the mean, then divide by the standard deviation, to turn x into a standard Normal z:

$$\begin{aligned} x &< 0 \\ \frac{x - 12}{16.5} &< \frac{0 - 12}{16.5} \\ z &< -0.73 \end{aligned}$$

Note how $x = 0$ is not always $z = -$

Figure 1.24 shows the standard Normal curve with the area of interest shaded.

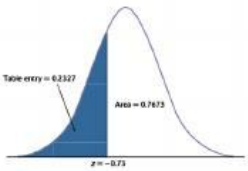


FIGURE 1.24 Areas under the standard Normal curve for Example 1.20.

3. *Use the table.* From Table A, we see that the proportion of observations less than -0.73 is 0.2327. The market is down on an annual basis about 23.27% of the time. Notice that the area to the right of -0.73 is $1 - 0.2327 = 0.7673$. This indicates that the market is up about 76.73% of the time.

◦ Assessing the normality of the data :-

⇒ Normal distributions provide good models about real world data, ex: milk per gallon of 2001 model year vehicles, average payroll of major league baseball teams, and estimate unemployment rates.

⇒ The distribution of some other common variables are usually skewed and therefore distinctly non normal, ex: Personal incomes, gross sales of businesses firms, and the service lifetime of mechanical or electrical components.

A histogram or stemplots can reveal distinctly non normal features of a distributions such as outliers, pronounced skewness, or gaps and clusters.

⇒ However, we need a more sensitive way to judge the adequacy of a normal model.

↳ The most useful tool to assessing the Normality is another graph is the normal quantile plot.

↳ It's not feasible to make normal quantile graph by hand, need a software to do it.

⇒ to calculate normal quantile plot :-

1) Arrange the observed data values from smallest to largest.

↳ Record what percentile of the data each value is occupying.

2) Do normal distributions calculations to find z-scores at these same percentiles.

↳ For example if the percentage is 5% is the percentile for a data value, the z-score from table is is $z = -1.645$.

3) Plot each data point x against the corresponding z.

↳ If the points are close to standard Normal, the plotted points will lie close to the 45-degree line $x=z$.

↳ If the data distribution is close to any Normal distribution, the points will lie close to a some straight line (with any slope m).

Standardization is a linear transformation that can change the slope and intercept of the line in our plot, but cannot turn it into a curved pattern.

Outliers in normal quantile plots appear far away from the overall pattern of the plot.

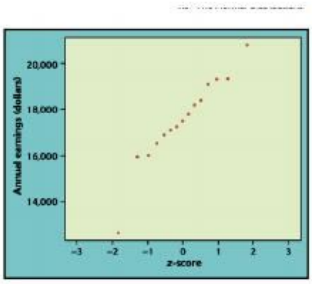
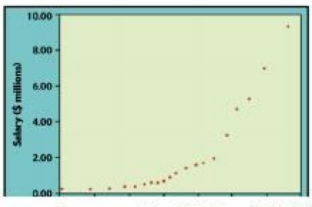


FIGURE 1.27 Normal quantile plot of the earnings of 15 black female hourly workers at National Bank, for Example 1.23. This distribution is roughly Normal except for one low outlier.



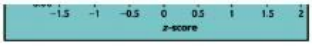


FIGURE 1.28 Normal quantile plot of the salaries of Cincinnati Reds players on opening day of the 2000 season, for Example 1.24. This distribution is skewed to the right.